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Seasonal population changes and malaria transmission potential of *Anopheles pharoensis* and the minor anophelines in Mwea Irrigation Scheme, Kenya

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A study in 1984 and 1985 showed that *Anopheles gambiae* s.l. and *An. pharoensis* were the major anophelines in Mwea Irrigation Scheme, Kenya, constituting 83.86% and 15.69% of the catch respectively. Four minor species made up the remaining 0.45%. The irrigation phase of the rice cultivation cycle in August, which linked the flooding effects of the two rainy seasons, resulted in major population increases of *An. pharoensis* and enabled continuous breeding for up to 9 months per year. The average of mean monthly proportions of unfed, bloodfed, and gravid females was 26.6, 58.8, and 14.6% respectively. The *Plasmodium falciparum* sporozoite rates for *An. pharoensis* were 1.3% by ELISA and 0.68% by dissection, while those for *An. funestus* were 1.7% by ELISA and 1.25% by dissection. *An. pharoensis* can contribute to the epidemiology of Malaria in the Mwea area.

Key words: *Anopheles pharoensis*; *Anopheles gambiae* s.l.; Mwea Irrigation Scheme; Seasonal variation; Gonotrophic condition; Sporozoite rates; Anophelines, rare

Introduction

Anopheles pharoensis Theobald is a widely spread mosquito in Africa, whose range extends from Egypt to South Africa, and from Mauritania to Somalia (Gillies and de Meillon, 1968). In East Africa, research efforts against the major malaria vectors of the *An. gambiae* Giles complex have largely been at the expense of barely any interest in *An. pharoensis*. With the recent increase in large scale irrigation farming however, the importance of other anophelines has increased, particularly with regard to the general lack of ecological information and disease transmission potential.

During 1984 and 1985, a survey of the mosquito population was carried out in the Mwea Rice Irrigation Scheme, Central Kenya. This report presents the seasonal dynamics and other ecological notes on *An. pharoensis* and other rare anophelines.

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Materials and Methods

The Mwea Rice Irrigation Scheme lies 120 km north of Nairobi in a riverine plain a few kilometres south-east of Mount Kenya, at an altitude of about 1200 m. A full description of the area is given by Mukiama (1987). The meteorological data are shown in Table 1.

The four villages selected for study in the scheme were Karima, Thiba, Rurumi and Mucii-wa-Urata (M.W.U.); consisting of 129, 133, 94 and 12 housing units respectively. The first three villages are within the irrigated zone, while the fourth one is peripheral.

In each village, mosquitoes were regularly collected on a fortnightly basis, in two structurally typical houses which served as permanent sampling stations. The key consideration in the selection of a particular house was the permission of the owner. The houses at Karima and Thiba were made of mud walls and corrugated iron roofs, those at Rurumi were made of concrete walls and corrugated iron roofs, while those at M.W.U. consisted of mud walls and grass thatched roofs. All invariably had large

TABLE 1

The meteorological data of Mwea Irrigation Scheme, 1984–1985

Month/year	Mean relative humidity (%)	Total rainfall (mm)	Mean temperature (°C)		Ricefield condition[a]
			Max.	Min.	
Jan 1984	48.5	5.8	29.9	14.4	NF
Feb	54.4	0[b]	31.6	14.3	NF
Mar	45.8	11.4	31.8	17.3	NF
Apr	56.8	144.1	30.9	18.9	PF
May	53.5	8.9	29.1	18.2	F
Jun	56.6	3.9	–	16.2	F
Jul	61.5	4.6	–	16.5	F
Aug	61.0	3.7	–	16.4	F
Sep	51.0	19.8	–	16.7	F
Oct	61.9	219.6	–	17.7	F
Nov	68.6	147.5	–	16.9	F
Dec	62.4	42.3	26.4	15.5	PF
Jan 1985	52.7	10.6	29.4	14.5	NF
Feb	48.9	68.5	30.2	17.2	NF
Mar	53.6	69.0	31.0	16.9	NF
Apr	66.1	475.5[b]	27.2	16.6	PF
May	67.5	54.3	26.1	17.5	F
Jun	66.8	0.9	24.9	15.9	F
Jul	62.0	2.4	24.7	15.8	F
Aug	56.0	6.0	24.9	15.7	F
Sep	51.9	1.5	29.0	17.0	F
Oct	53.8	47.2	29.2	17.0	F
Nov	67.8	136.4	27.4	17.3	F
Dec	60.5	33.2	28.0	14.9	PF
Monthly means	57.9	63.2	28.4	16.5	

[a]Key: F, flooded; PF, partially flooded; NF, not flooded.
[b]Indicates range.

eaves between the walls and the roofs for ventilation. In addition to people, chickens, sheep and goats occasionally stayed indoors overnight, while cattle and donkeys were penned in open enclosures between houses; although roofed cowsheds were present in a few homesteads.

Daytime indoor-resting mosquitoes were sampled by pyrethrum spray sheet collections between 9:00 and 11:00 h using a local commercial preparation 'Pyrethrol'. Overnight catches by CDC miniature light traps were made between 18:30 h and 6:30 h the next day. Mosquitoes leaving houses were sampled by the use of window exit traps, which were installed at 18:30 h and removed at 6:30 h the next day. A detailed discussion of these methods is given by Service (1976, 1977).

All mosquitoes were later identified in the laboratory by ordinary taxonomic characters described by Gillies and de Meillon (1968).

For each village, total numbers of a mosquito species collected per month were obtained by addition of fortnightly collections. The mean catch per month was then obtained by dividing the total by the number of catches made per month. These monthly means were pooled to give the mean catch per month for the whole area.

Gonotrophic condition of females was determined by simple dissecting microscope examination of females pre-treated with Carnoy's fixative. Sporozoite rates were determined by manual dissections of salivary glands, and an enzyme-linked immunosorbent assay (ELISA) for *Plasmodium falciparum* which differed from that described by Beier et al. (1987) only in the following minor details: (i) individual specimens were ground in blocking buffer with Non-indet P-40 using motorised wooden applicators, (ii) following incubation and the addition of the peroxidase substrate, the plates were read using the ELISA reader at 414 nm after 30 min, and (iii) mosquitoes were considered infected if ELISA absorbance values exceeded the mean multiplied by 2 of the control mosquitoes on the same plate.

Results

Table 2 shows the total numbers and percentages (i.e. combined pyrethrum spray and light trap collections) of the anophelines in Mwea. *An. gambiae* s.l., the most abundant species is discussed elsewhere (Mukiama and Mwangi, in press). *An. pharoensis* was a common mosquito, while *An. funestus* Giles and *An. pretoriensis* Theobald may be considered rare species. *An. maculipalpis* Giles and *An. coustani* Laveran however were very rare, both constituting less than 0.1% of the total catch.

TABLE 2

The anopheline species in Mwea Irrigation Scheme, total for 1984–1985

Species	Number	Percentage
An. gambiae s.l.	15374	83.86
An. pharoensis	2876	15.69
An. funestus	49	0.27
An. pretoriensis	23	0.12
An. maculipalpis	9	0.05
An. coustani	2	0.01
Totals	18333	100.00

Table 3 shows the species composition of both daytime and overnight catches in different villages, including *An. gambiae* s.l. *An. pharoensis* was consistently common in all villages, and for example, accounted for 81.2% and 42.4% of the daytime and overnight catches respectively at Karima village. Both *An. funestus* and *An. maculipalpis* were caught resting indoors during the day more frequently than *An. pretoriensis*, which appeared to prefer visiting houses at night.

An. pharoensis

Overnight forays into houses were much more evident in *An. pharoensis*, where light trap catches were more than two-fold the daytime catches (Fig. 1). The correlation ($r = 0.39$) between the two catches is not significant ($p > 0.05$, $n = 20$, 18 d.f.).

Exit catches and the combined total catch are significantly correlated ($r = 0.78$, $p < 0.05$, $n = 12$, 10 d.f.), indicating similar changes in the same direction (Fig. 2). The combined total catch and the rainfall data for the area however did not correlate ($r = 0.05$, $p \gg 0.05$, $n = 20$, 18 d.f.). This indicates that actual precipitation is not in the short term effective in adult mosquito production, since time must elapse for water bodies to form to facilitate effective larval breeding.

A summary of the data on the gonotrophic condition of *An. pharoensis*, showing the total mean monthly percentage composition of catches by pyrethrum spray, light-traps and exit traps; together with the observed variability is given in Table 4.

The sporozoite rates obtained were 1.3% by ELISA (1 out of 77 specimens) and 0.68% by dissection (1 out of 146 specimens).

TABLE 3

The composition of daytime and overnight samples of anopheline mosquitoes from all four villages in Mwea (total 1984–1985)

Species	Village				Total	Percentage
	Thiba	Rurumi	M.W.U	Karima		
Daytime						
An. gambiae s.l.	804	1449	306	4996	7555	89.14
An. pharoensis	45	103	14	701	863	10.18
An. funestus	7	37	0	0	44	0.52
An. pretoriensis	0	0	3	0	3	0.04
An. maculipalpis	0	3	5	0	8	0.10
An. coustani	0	2	0	0	2	0.02
Total	856	1594	328	5697	8475	100.00
Overnight						
An. gambiae s.l.	4125	1384	647	1946	8102	79.80
An. pharoensis	474	653	39	859	2025	19.94
An. funestus	5	0	0	0	5	0.05
An. pretoriensis	20	0	0	0	20	0.20
An. maculipalpis	0	1	0	0	1	0.01
An. coustani	0	0	0	0	0	0
Totals	4624	2038	686	2805	10153	100.00

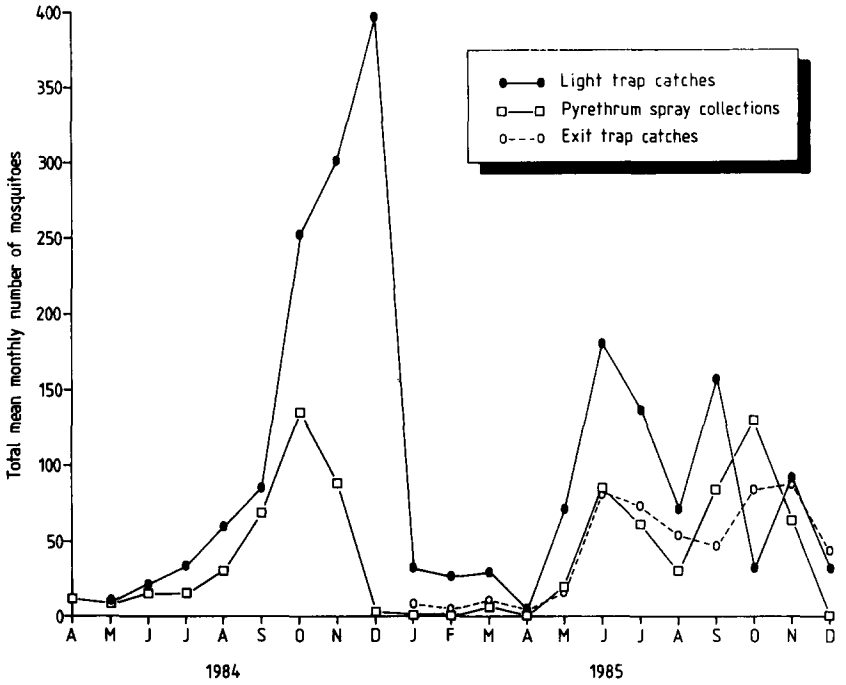


Fig. 1. Variations in the mean monthly catches of *An. pharoensis* at all sampling locations by different methods.

TABLE 4

A summary of the gonotrophic condition by catch method of female *An. pharoensis* caught in Mwea Irrigation Scheme, 1984–1985

Catch	Gonotrophic condition (%)			(*n*)
	Unfed	Fed	Gravid	
Pyrethrum spray	6.9[b]	77.6[b]	15.5	(595)
Light-traps	27.4	53.3	19.3	(1622)
Exit traps	45.6[b]	45.6	8.8[b]	(342)
(*n*)	(641)	(1482)	(436)	(2559)
Average of monthly means	26.6	58.8	14.6	
S.E.[a]	9.1	7.9	2.5	
95% confidence limits:	8.8–44.4	43.3–74.3	9.7–19.5	

[a]S.E, standard error.
[b]Deviations significant, $p < 0.05$.

An. funestus

All specimens were collected in the southerly villages of Thiba and Karima (Table 3). Over 90% of them were caught in 1984. An examination of the females caught during daytime showed that 71.4% were bloodfed, 23.8% gravid and 4.8% unfed. All four females caught overnight were bloodfed. No specimens were caught by exit traps. A

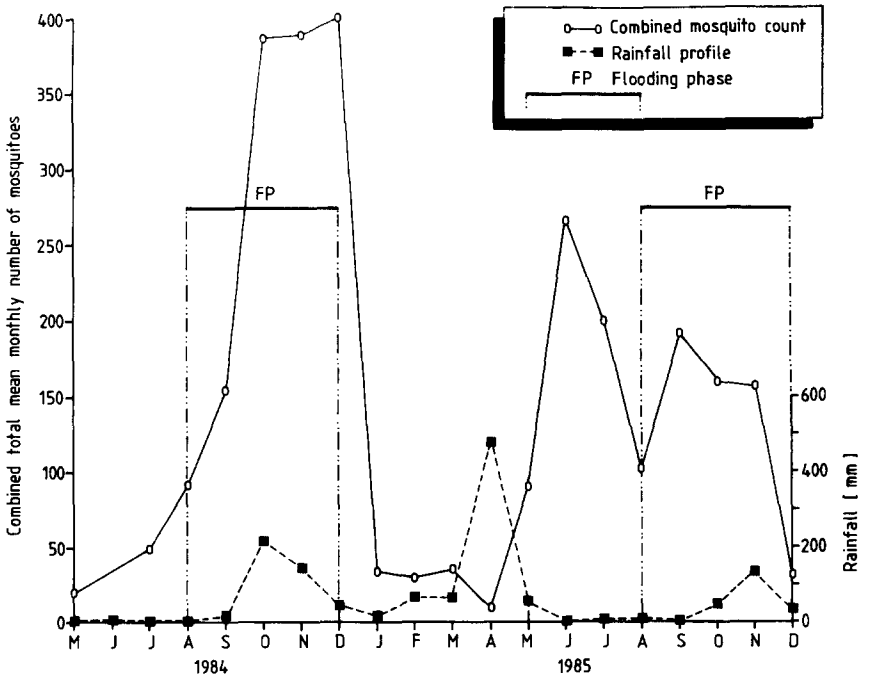


Fig. 2. Variations in the combined mean (pyrethrum spray and light trap) catches of *An. pharoensis* at all sampling locations per month shown against the rainfall profile of Mwea and the flooding phase of the rice cycle.

later survey of this species in the area during the long rainy season of 1986 (April–June) showed a sporozoite rating of 1.7% by ELISA (6 out of 344 specimens) and 1.25% by dissection (1 out of 80 specimens).

Discussion

The population changes of *An. pharoensis* observed in Mwea, particularly the peaks (Fig. 2), demonstrate the combined effects of irrigation and rainfall versus the complete absence of surface water in the rice fields during the dry season. The long rainy season usually starts late March, but is normally not fully effective until mid-April to early May, when widespread flooding occurs. The fields are then completely waterlogged, and this may be sustained until mid-July, when the increasing effects of evaporation considerably reduce the flooded areas. Irrigation in early August however soon restores the lost surface water, thus maintaining perfect mosquito breeding conditions. The irrigation activity eventually merges with the short rainy season, which commences in October. This situation therefore may enable continuous breeding for up to nine months per year, since the crucial linkage between the rainy seasons is provided by irrigation.

In addition to rice fields (Khamala, 1971; Surtees, 1971), *An. pharoensis* has also been found breeding in river fringes and swamps (Bevan, 1937; Lewis, 1948), lake shores and papyrus swamps (Bevan, 1937; Haddow, 1942; Garnham, 1945; Goma, 1960), inland swamps and reservoirs (Lewis, 1948, 1958; Goma, 1960), irrigated land

and irrigation ditches (Lewis, 1948; El Safi and Haridi, 1986), muddy puddles and open ponds (Khamala, 1971).

Significant deviations (Table 4) were observed in pyrethrum spray catches and exit trap catches. In the former, the proportion of unfed females was lower than expected, while that of bloodfed females was higher than expected. In the latter, the proportion of unfed ones was higher than expected, while that of gravids was lower than expected. These observations indicate that fed females may often be found resting indoors, while unfed and gravid ones tend to rest outdoors. This inference is also supported by the trapping of a high percentage (45.6%) of bloodfed females by exit window traps.

Haddow (1942) and Lewis (1948) have expressed similar views on the exophilic nature of *An. pharoensis*, with the observations that females enter houses at night mainly for feeding purposes, only to retreat before daybreak. This general nocturnal activity of this species is widely recognised (Kerr, 1933; Haddow, 1942; Hanney, 1960; Service, 1963; Haddow and Ssenkubuge, 1974). On the contrary, daytime activity has been observed to be mainly limited to resting outdoors among vegetation for the majority, with only a small proportion staying indoors (Gibbins, 1932; Lewis, 1948, 1958; Gillet, 1971).

An. pharoensis was found present in Mwea during the entire duration of the study (Fig. 2), which is consistent with its natural range in the Ethiopian zoogeographical region (Gillies and de Meillon, 1968). These observations do not suggest any migratory or swarming behaviour observed in the species elsewhere (Garrett-Jones, 1950; Haddow et al., 1951; Haddow and Ssenkubuge, 1974).

Lewis (1958) noted that *An. pharoensis* was the principal man-biting anopheline outdoors in the Kosti area of Sudan in the evening hours, but rarely bit indoors. Lewis (1948) also demonstrated its preference for man over a calf in sharp contrast to Haddow (1942) who showed overwhelming preference for the latter. However, Garrett-Jones et al. (1980) recently reported an indoor sample of *An. pharoensis* from Ethiopia showing a man-fed index of 94%. Boreham and Port (1982) caught 2 human blood positive specimens in bed nets that had been occupied the previous night in the Gambia. Available information therefore indicates that, although samples from different areas demonstrate varying levels of human or bovid blood preference, *An. pharoensis* certainly bites man, and its medical importance should therefore not be underestimated. Further investigations are therefore required to establish the identity of the high bloodfeeding indices reported in this study.

Reports on the vectorial capacity of *An. pharoensis* indicate wide differences in different populations. According to Gillies and de Meillon (1968), *An. pharoensis* is a proven vector of malaria in Egypt and some countries in West Africa; but results from other areas of Africa are rather inconclusive. The reported sporozoite rates are much lower than those observed in the *An. gambiae* complex or *An. funestus*. Garnham (1945) reacted to the observed variation in sporozoite infections in different populations of *An. pharoensis* by suggesting that varieties of the species with different vectorial capacities may be present. This view is particularly interesting following recent cytogenetic observations by Miles et al. (1983) indicating that the taxon *An. pharoensis* may be a complex of two species. The present demonstration of *P. falciparum* sporozoites in *An. pharoensis* cannot be fully evaluated until larger samples from the area are systematically examined over a longer period.

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